

SCIENCE CENTRE OBSERVATORY · SINGAPORE



SCOB Night-Sky Dashboard

Volunteer & Docent User Guide

A single-file web app that plans every Friday-night session — Moon, planets, deep-sky targets, weather and the run-sheet, all computed on-device.

Open: [scob-dashboard-v3.html](#)

What the dashboard is

It is a **single web page** you open in any browser. Everything — Moon phase, planet altitudes, deep-sky target rankings, the sky map, sunset and twilight times — is **calculated on the device**, so it keeps working with no internet. Only the **live weather** needs a connection.

One canonical file to open or host: [scob-dashboard-v3.html](#). The older scob-dashboard.html and main.html are now just redirect stubs; everything superseded lives in archive/.

Offline-first

Works in the dome with no signal. Weather is the only online piece.

Built for SCOB

Every verdict is tuned to our fleet, our site and our 7:30–10 pm window.

One file

Nothing to install; open it, or add it to the home screen as an app.

Quick start — three steps

1

Open the page

Double-click scob-dashboard-v3.html, or open the hosted URL. On phones, tap Share → Add to Home Screen to install it as an app (see INSTALL-as-app.md).

2

It defaults to the session

The date auto-sets to the next Friday, 8:30 pm SGT, with a live countdown. Use ● Now or ⤴ Next Fri to jump, or the slider to scrub the night.

3

Read top-to-bottom

Start at Tonight at a glance for the go/no-go call, then work down: sky map, Moon, planets, deep-sky, weather, events, run-sheet.

Tip: tap ☰ Menu (top-left) for all the guides & tools; your view (Desktop / Mobile) & night mode are remembered.

Date & time planning

The bar at the top of the page drives everything below it. Change the moment and every card — altitudes, verdicts, weather — recomputes instantly.

Date & time pickers

Choose any night and any time; the page flags whether it is a Friday session night.

Time slider

Drag to scrub across the whole evening and watch objects rise and set.

● Now

Snap to the current moment — handy while running the session live.

↶ Next Fri

Jump straight back to the next Friday, 8:30 pm default.

Live countdown

Shows time remaining to the chosen moment.

Friday flag

A clear marker when the chosen date is our public-session night.

Tonight at a glance

The first card gives a one-line **go / no-go verdict** that combines the live forecast with sky darkness — your fastest read on the night. Below it, quick chips summarise the essentials.

Sunset

Fully dark

Sunrise

Moon % + altitude

Best planet

Top deep-sky target

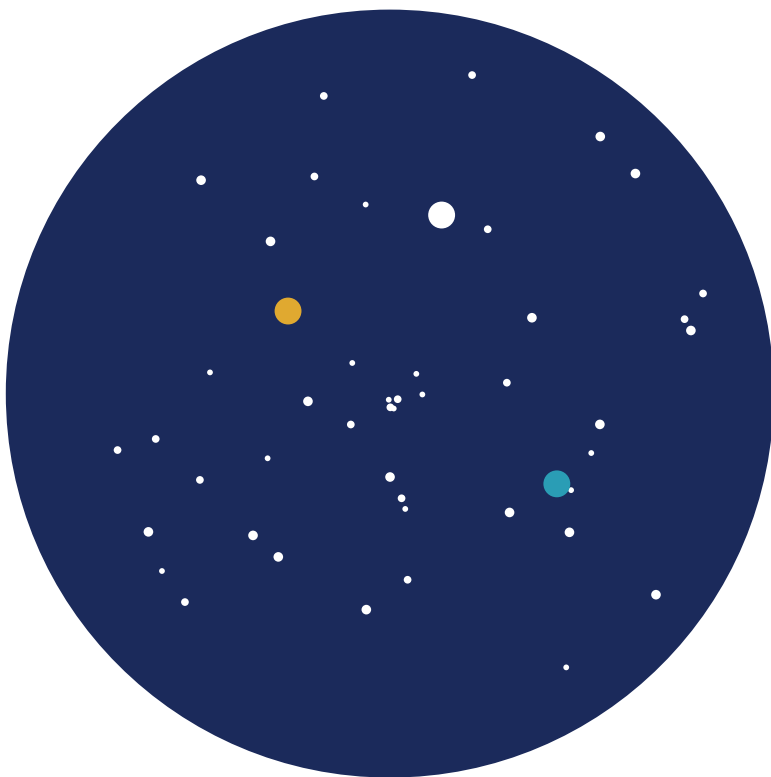
Tonight's event

Best satellite pass

How to use it: glance at the verdict when you arrive, then let the chips point you to the highest-value targets for the conditions. If the Moon chip is bright and high, expect the deep-sky list to favour clusters and double stars over faint fuzzies.

The sky map

N (looking up · E to the left)

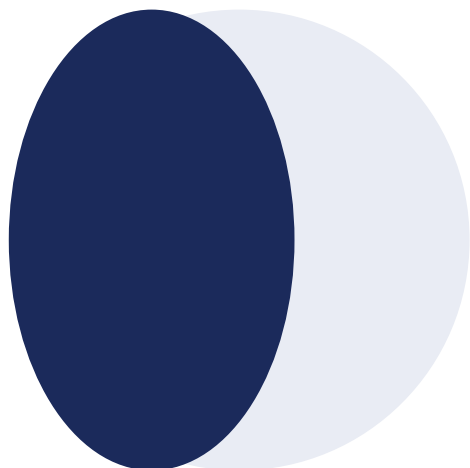


An all-sky “looking up” chart

North is at the top and East to the left, matching the view when you tip your head back.

- Plots planets, the Moon, the Sun and the ranked deep-sky targets for the chosen moment.
- 17 constellation stick figures — tap the map to toggle them on or off.
- Hover or tap any object for an illustrated tooltip explaining what it is.
- Great for pointing visitors to what's actually overhead right now.

The Moon card



Illuminated disc + rise/set

Everything you'll tell a visitor about tonight's Moon:



Phase & % illuminated

Named phase with a graphical disc showing exactly how lit it is.



Rise & set times

So you know when it clears the horizon or leaves the sky.



Moonlight impact

Plain-language notes on how much it will wash out faint targets.

Deep link: the card opens the dedicated Tonight on the Moon page (slide 15) for terminator detail.

Sun, Moon & the planets

A single table lists the Sun, Moon and all seven planets for the chosen moment, each with a plain visibility verdict.

Body	Altitude	Direction	Mag	Verdict
Jupiter	58°	SE	-2.3	Scope — superb
Saturn	34°	E	0.9	Scope — rings
Venus	12°	W	-4.0	Twilight, low
Mars	-6°	—	1.4	Below horizon

Illustrative values

Reading the verdict

Naked eye: bright enough to just point at

Binoculars: worth a quick handheld look

Scope: put it in a dome or CPC

Twilight: catch it early before it sets

Below horizon: not up — skip it

Deep-sky targets — two modes

A Messier + Caldwell list, ranked easiest-to-hardest, each with magnitude, compass direction and **Moon separation**. Toggle between two verdict modes — your choice is remembered.



Visual mode

Eyepiece verdicts

- Excellent / Good / Low ratings by altitude
- Washed-out flag under a bright Moon
- The default for normal visual nights



Seestar S30 Pro mode

Imaging verdicts

- Accounts for the dual-band H α /OIII filter
- Emission nebulae stay imageable under moonlight
- Broadband galaxies flagged for dark weeks

Moon separation is the key column on bright nights — small numbers mean the target sits near the Moon and will be swamped. v3.10: the list is now ranked easiest-to-hardest and previews the top 5 with a “Show all” toggle; verdicts account for Jurong’s city sky (a Bortle selector) and each row shows its observing window & best time.

Weather — the only online card

Auto-loads a live forecast (Open-Meteo) for the chosen date & time and returns a **GO / CAUTION / NO-GO** verdict. The button just refreshes it. A seeing & transparency forecast (7Timer!, with an offline estimate as fallback) shows how steady and clear the air will be — the real decider for planets & doubles.

Forecast detail

Cloud, rain chance, temp, humidity, wind.

Evening cloud timeline

Hourly bars 6–11 pm; clearest window highlighted.

MSS 2-hr Nowcast

In-page Jurong East nowcast (data.gov.sg).

Live rain radar

MSS 50-km radar image, 5-min timestamps.

Haze / PSI

Live NEA west-region PSI with a transparency note.

Monsoon guidance

Seasonal notes for dates beyond the ~16-day window.

On the night: quick links to the MSS 2-hr Nowcast and SCOB Facebook back up the forecast for the final call.

Special events calendar

A verified 2026–27 calendar of **meteor showers, eclipses, oppositions, elongations and supermoons**. Events in the three weeks after the chosen date are flagged; anything not visible from Singapore is dimmed. Fully offline — the night's own event also shows as a glance chip.



Meteor showers

Peak dates & rates



Eclipses

Solar & lunar, SG-visible flagged



Oppositions

Best nights for the outer planets



Supermoons

Closest full Moons of the year

Satellite passes

Evening (7–11 pm) pass predictions for the **ISS, Tiangong and Hubble**, computed on-device from CelesTrak TLEs via satellite.js. The TLEs cache in the browser and refresh every ~3 days when online. The best visible pass appears as a glance chip.



Rise • peak • set

Exact times so you can line up the crowd in advance.



Direction & max altitude

Where to point and how high it climbs.



Sunlit vs shadow

Flags whether the pass is actually visible or in Earth's shadow.

The session run-sheet

An auto-generated, per-instrument plan for **7:30 · 8:30 · 9:30 pm** — so each scope has a job all evening.



Deck CPC / C6

The Moon — easy to find, quick to reset for the queue.



40 cm & 11" domes

The best planets — Jupiter and Saturn at high power.



40 cm + 6" apo

Globulars & clusters that reward aperture.



Seestar S30

Emission nebulae, using the dual-band filter.

Why it matters: hand each volunteer their instrument's line and the whole dome runs itself — targets are matched to aperture and to how bright the Moon is.

The printable briefing



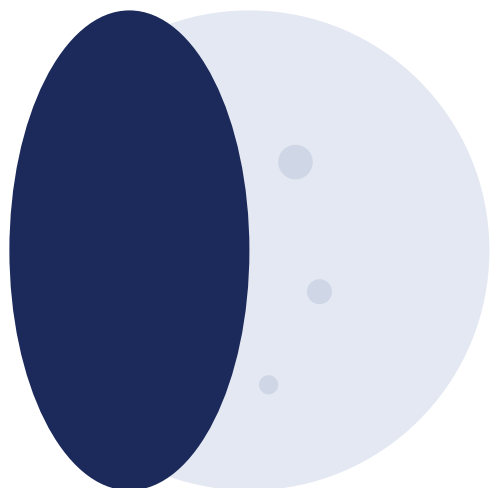
Print or save as PDF

One tap builds a print-friendly weekly briefing

- Glance verdict, weather, Moon and planets
- Top-10 target list
- Upcoming events & satellite passes
- The full session run-sheet
- The sky-map SVG, embedded in the page
- Data sources credited in the footer

Print it, or save as PDF straight from the browser — the weekly handout for the team, produced in one click.

Tonight on the Moon



Computed terminator

A dedicated page for lunar nights

- Drawn Moon disc with the computed sunrise/sunset line, maria and craters for 8:30 pm.
- “On the terminator tonight” — craters, rilles and ranges within $\sim 13^\circ$ of the line, ranked with docent one-liners and scope tips (35-feature IAU/USGS catalogue, incl. Tranquility Base).
- Full-Moon nights switch to the ray systems story (Tycho, Copernicus, Kepler, Aristarchus).
- New-Moon nights point back to deep-sky; earthshine tip on thin crescents.
- Phase, age, colongitude, rise/set and an up-during-session verdict.

Jupiter & Saturn at the eyepiece

Extra detail for the two showpiece planets, computed for the exact time you're observing.



Great Red Spot

Meridian-crossing (transit) times for the evening — know when the GRS faces us. Uses a Meeus CM series plus the JUPOS drift trend.



Galilean moons

Io, Europa, Ganymede and Callisto drawn to scale for the chosen time, flagging transits and occultations.



Saturn ring tilt

Computed on-device with an opening / edge-on interpretation.

Double stars & conjunctions



Double & carbon stars

18 showpieces that shrug off moonlight — the go-to list when the Moon washes out the fuzzies.

- Albireo, α Centauri, Acrux
- The Double-Double (ϵ Lyrae)
- La Superba, DY Crucis (carbon reds)
- Mags, separations, colours listed
- Altitude-ranked verdicts



Conjunction finder

Finds close pairings on-device for 8:30 pm over the three weeks after the chosen date — no longer limited to the static events calendar.

- Moon–planet pairings within 5°
- Planet–planet pairings within 2.5°
- Tonight's pairing shows as a glance chip

Telescope & optics guide

A cross-linked teaching page — great for explaining the fleet to visitors and new volunteers.

Five optical designs

Animated light paths: refractor, Newtonian, classical Cassegrain, Schmidt-Cass and Maksutov — each notes which SCOB scope uses it.

Four mount types

Animated alt-az GoTo fork (CPC), Dobsonian, German equatorial and the English yoke on the 40 cm main scope.

DSO tracking explainer

Equatorial vs alt-az GoTo, field rotation, and the Singapore quirk of a celestial pole on the horizon.

Magnification calculator

Focal length ÷ eyepiece, with one-tap presets for the whole SCOB fleet, f-ratio, exit pupil and useful-magnification warnings.

Tracking & alignment methods

The Celestron alignment reference — how GoTo mounts find and follow targets. Note: SCOB's CPC mounts have **no GPS module** — date, time and location are keyed into the hand control by hand.

SkyAlign

1-star align

2-star align

Auto 2-star

Solar System align

ASPA polar align

Also covers: sidereal / lunar / solar tracking rates and a side-by-side comparison of the SCOB mounts. Cross-linked with the telescope guide and cached for offline use.

The SCOB telescope fleet

The dashboard's verdicts and run-sheet are tuned to these instruments.



Main dome

40 cm (16") Cassegrain + 15 cm (6") apo

Best for: Planets, globulars, tight doubles



Second dome

11" + 7"

Best for: Bright planets and clusters for the queue



Deck / portable

Celestron CPC (8"–11") + C6 SCT

Best for: Moon and quick crowd-pleasers

App conveniences



Night mode

Red-tinted, dark-adaptation friendly. Switches on automatically after sunset until you override it; your manual choice is remembered.



Two layouts, one file

Desktop and mobile auto-detected, with a manual switcher. The mobile view has a slim sticky header for the date/time/slider.



Install as an app

Add to the home screen as a PWA (manifest, service worker, icons). Once cached, every page works offline — see [INSTALL-as-app.md](#).

The latest upgrades

Versions 3.7–3.10 sharpen the calls you make on the night — all computed on-device, weather aside. Here is what changed.

Seeing & transparency

A 7Timer! astro forecast (with an offline estimate as fallback) rates how steady and clear the air will be — the real decider for planetary detail.

City-sky realism

A Bortle selector (default 8, typical Jurong) trims faint galaxies & nebulae the light-polluted sky won't actually show.

Observing windows

Each deep-sky target shows when it clears 25° and its best time. The list is ranked easiest-to-hardest, with the top 5 shown by default.

Live “Now” mode

A full-screen, red-mode, auto-refreshing view: the best target per instrument at this minute. Ideal at the eyepiece.

Data-health strip

Surfaces forecast age, orbital-element (TLE) age and the Jupiter GRS review date, so stale inputs are visible at a glance.

Tap to locate (v3.10)

Tap any object — in the target list, the Solar-System table or the double-star list — to flash a pulsing ring on the sky map, so anyone can find it fast.

New pages, kiosk & tracks

New companion pages plus a dashboard add-on — all computed on-device and cached for offline use.

Tonight's Tour

A printable docent script — what to show the public in order, with talking points and the right SCOB scope for each object.

Point-your-phone finder

Pick a target and the phone's compass steers you to it with a live arrow; a text direction & altitude fallback covers phones without sensors.

Eyepiece & field of view

Choose a scope + eyepiece to see magnification, true field and exit pupil, with the target framed to scale so you know what will fit.

Satellite tracks on the map

An optional overlay draws tonight's ISS, Tiangong & Hubble pass arcs on the sky map — a toggle, off by default so it never clutters.

Visitor kiosk & QR

A full-screen venue display — what's up now in plain language, plus a QR code visitors scan to open the finder on their own phone.

All on-device

No accounts and no internet needed once installed — one shared astronomy engine (astro-core.js) powers the tour & finder.

Before the gates open

- 1 Open the page — it's already on next Friday, 8:30 pm.
- 2 Check Tonight at a glance for the go/no-go verdict.
- 3 Scan the weather card: cloud timeline, radar, PSI.
- 4 Note the top deep-sky targets and the best planet.
- 5 Read the run-sheet — hand each dome its line.
- 6 Tap the Briefing button and print the handout for the team.